

Efficacy and Safety of Two Strengths of Sitagliptin, Metformin and Glimepiride in Type 2 Diabetes Mellitus: A Subgroup Analysis of Phase 3 Study in India

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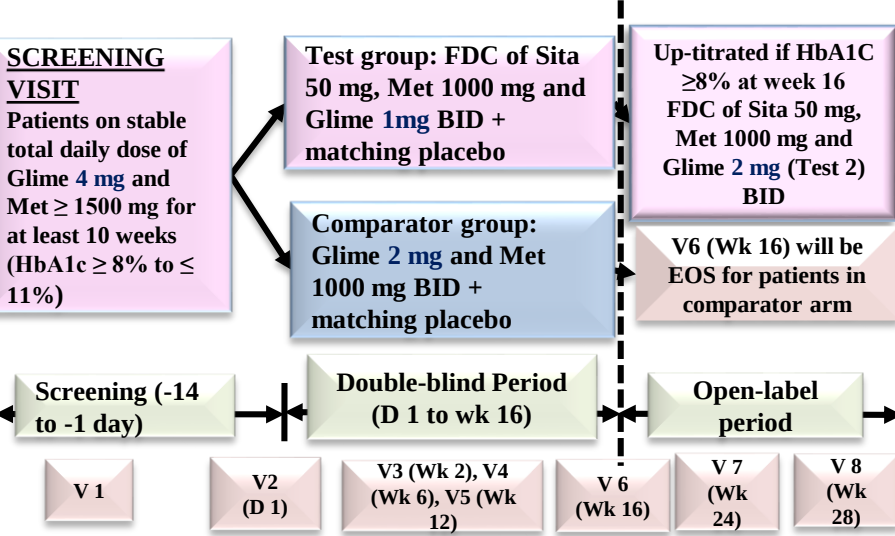
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OBJECTIVE

This is a subgroup analysis of the phase 3 clinical study conducted to evaluate the efficacy and safety of two fixed-dose combinations (FDCs) of Sitagliptin+Glimepiride+Metformin (50+1+1000 mg, 50+2+1000 mg) twice daily (BID) in type 2 diabetes mellitus (T2DM)

METHODS

➤ This is a subgroup analysis of a phase 3, randomized, double-blind, double-dummy, active controlled study (16 weeks) followed by an open-label, single-arm study (12 weeks) [CTRI/2021/11/038169]



BID: Twice daily; D: Day; EOS: End of study; FDC: Fixed dose combination; Glime: Glimepiride; HbA1c: Glycated hemoglobin; Met: Metformin; Sita: Sitagliptin; V: Visit; Wk: Week

➤ The glycemic parameters and safety profile of all patients were evaluated from Baseline to Weeks 24 and 28 and from Week 16 to Weeks 24 and 28

RESULTS

Baseline Demographics:

- A total of 182 patients included in this subgroup analysis received Test 1 from Baseline to Week 16.
- At Week 16, 52 patients with HbA1c ≥ 8% were uptitrated to Test 2, and remaining 130 patients continued to receive Test 1 till Week 28
- The mean age was 47.65 ± 8.47 years and the mean BMI was 25.63 ± 3.55 Kg/m²

Baseline Characteristics:

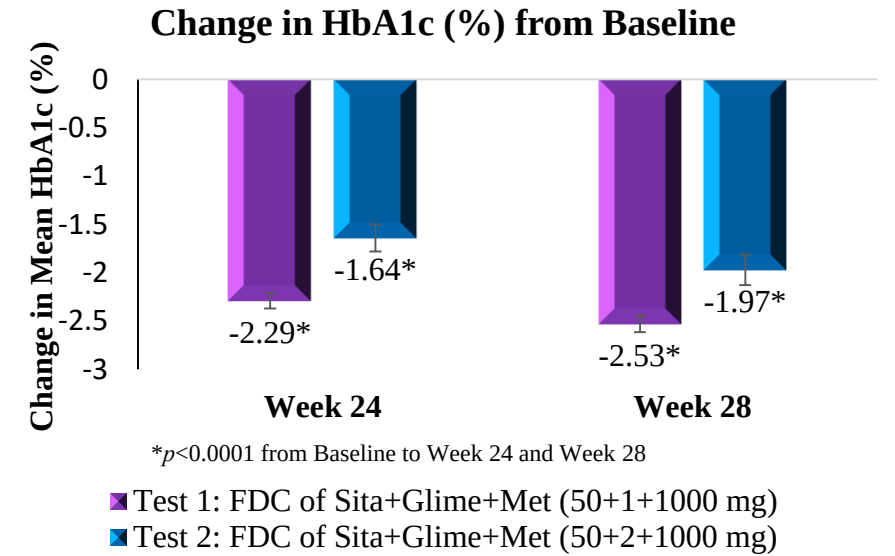
The glycemic parameters of the treatment arms at Baseline and Week 16

Parameters	Visit	FDC of Sita+Glime+Met (50+1+1000 mg) (Test 1) (N=130) Mean ± SD	FDC of Sita+Glime+Met (50+2+1000 mg) (Test 2) (N=52) Mean ± SD
HbA1c (%)	Baseline	8.97±0.61	9.64±0.83
	Week 16	6.93±0.75	8.51±0.64
FBG (mg/dL)	Baseline	176.3±34.41	177.7±46.23
	Week 16	129.8±25.62	139.1±37.80
PPBG (mg/dL)	Baseline	245.4±49.57	265.4±61.69
	Week 16	179.2±32.95	207.0±44.09

FBG: Fasting blood glucose; FDC: Fixed dose combination; Glime: Glimepiride; HbA1c: Glycated hemoglobin; Met: Metformin; PPBG: Postprandial blood glucose; Sita: Sitagliptin; SD: Standard deviation

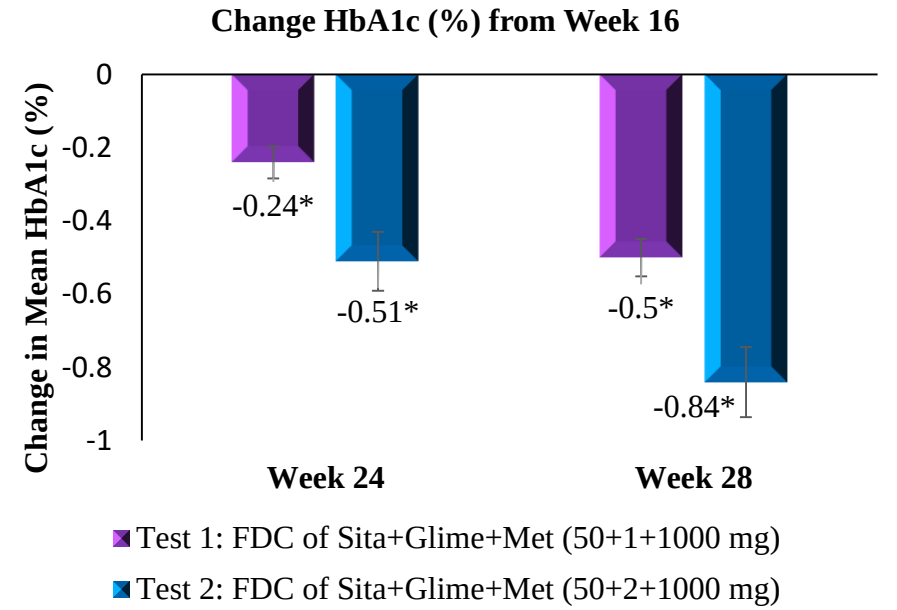
Change in Mean HbA1c from Baseline to Weeks 24 and 28:

Both Test 1 and Test 2 arms demonstrated significant reduction in mean HbA1c from Baseline to Weeks 24 and 28 ($p < 0.0001$ each)



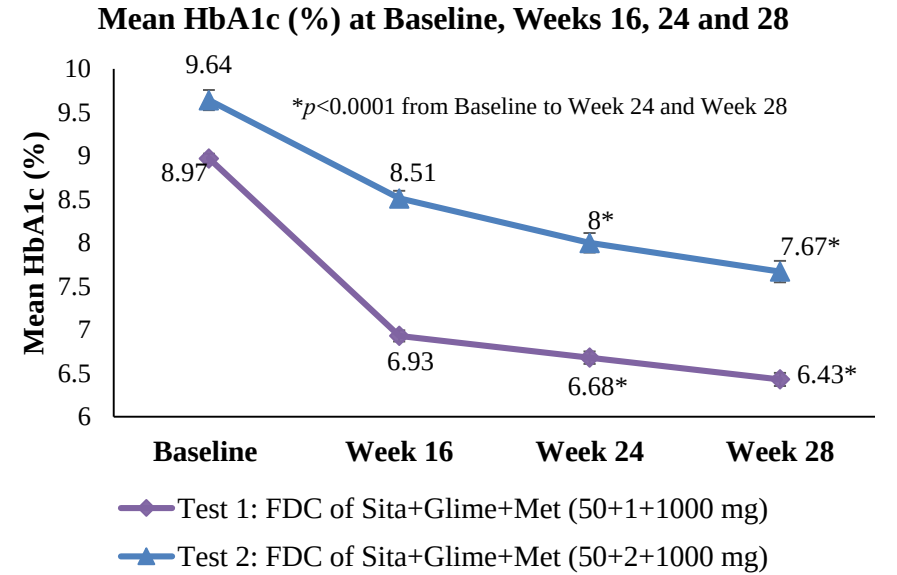
Change in Mean HbA1c from Week 16 to Weeks 24 and 28:

Both Test 1 and Test 2 arms demonstrated significant reduction in mean HbA1c from Week 16 to Weeks 24 and 28 ($p < 0.0001$ each)



Change in HbA1c (%) from Baseline and Week 16:

The mean HbA1c (%) of the uptitrated and non-uptitrated patients



Change in FBG and PPBG from Baseline and Week 16:

- Statistically significant reduction in both FBG and PPBG from Baseline to Week 24 and 28 were observed in both the Test arms ($p < 0.0001$ each)
- Statistically significant reduction in both FBG and PPBG from Week 16 to Week 24 and Week 28 were observed in Test 1 arm ($p < 0.0001$ each), and statistically significant reduction in FBG and PPBG from Week 16 to Week 28 was observed in Test 2 arm ($p < 0.05$ each)

Proportion of patients achieving HbA1c < 7.0%:

- Proportion of patients achieving HbA1c < 7.0% was 56.2% and 9.6% in Test 1 and Test 2 arms, respectively at Week 24
- Proportion of patients achieving HbA1c < 7.0% was 68.2% and 15.4% in Test 1 and Test 2 arms, respectively at Week 28

Adverse events – Overall summary:

- Overall 8 adverse events (unlikely related) were reported during Weeks 16 to 28, including one hypoglycaemia event (level 1) requiring no intervention
- Out of total 8 TEAEs in the test arm, 7 TEAEs occurred in patients in Test 1 arm and 1 TEAE occurred in patient in Test 2 arm
- All the TEAEs were “unlikely” related to the study treatments
- No SAEs, deaths, serious or life-threatening, and drug-related TEAEs were reported.

DISCUSSION AND CONCLUSION

- The results of the study indicated that the addition of sitagliptin to glimepiride treatment in patients with T2DM led to improved glycemic control till Week 28
- Patients who responded to low dose FDC (with glimepiride 1 mg) at Week 16 continued to demonstrate further significant improvement in glycemic parameters, including HbA1c, FBG and PPBG
- Patients who required uptitration (FDC with glimepiride 2 mg) at Week 16 demonstrated significant improvement in glycemic parameters till Week 28
- Both the FDCs were safe and well tolerated